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Design of intelligent vehicular and sensor communication network: a comprehensive survey

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ABSTRACT

This review paper provides a comprehensive review of Intelligent Vehicular and Sensor Communication Networks (IVSCN), focusing on the design details and recent advancements required to develop modern vehicular systems. It highlights the comprehensive sensor technologies such as Radar, Light Detection and Ranging (LiDAR), Cameras, and Ultrasonic sensors that aim to collect and synchronize data for vehicle operation and safety. The paper also critically assesses sophisticated network designs and architectures that strengthen vehicular communications. Further in-depth analysis is provided about vehicular network communication protocols and navigation systems, which can manage a multitude of traffic scenarios, including signal control, accidents, and congestion, which are invaluable for optimizing vehicle movement and ensuring road safety. Moreover, the review further covers practical applications and cases, demonstrating that these technologies have worked well, from providing sound traffic management systems to improving vehicle safety and environmental sustainability. The review also explores other Artificial Intelligence (AI) techniques used in fault-tolerant and intelligent communication methods, such as Kalman Filters, Particle Filters, and Fuzzy Logic Fusion. This literature review highlights the role of ongoing innovation in routing protocols, navigation systems, and data analytics; it thus has practical implications for both upcoming and well-known researchers.

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Introduction

The Intelligent Vehicular and Sensor Communication Network (IVSCN) forms the basis of key innovations within the transportation industry by incorporating cutting-edge technology with the smart features of intelligent transport systems (ITS) (Meng et al., 2024) and the advanced communication skills of vehicular ad-hoc networks (VANETs) (Ahmed & Cheelu, 2024). It is a moment of crossroads where both technological inventions have come together to open an era of connected driving that is ultimately aimed at providing travellers with an expanded level of safety on the road, limited traffic jams on highways, and environmental protection.

The journey of IVSCN began at the end of the 1990s with the creation of infrastructure, including vehicles that could communicate themselves and road infrastructure. The proposed system manages to accomplish this by creating road safety by sharing different attributes of vehicles, such as speed and direction. Thus, collisions are prevented. Dedicated Short-Range Communication

(DSRC) (Thompson & Kim, 2024) is one of the earliest technologies and has received much support from 802.11p, allowing prompt communication between vehicles and roadside units (RSUs). In the precedence of the new millennium, the development of cellular technologies gave birth to C-V2X (Cellular Vehicle-to-Everything) (Rayamajhi et al., 2020) using long-term evolution (LTE) and as a final stage of 5G networks (Jellid & Mazri, 2021; Petrov et al., 2021). This shift demonstrated significant advantages regarding the range, reliability, and capabilities, which increased the absence of DSRC limitations. In summary, sensor technology developments, notably from radar, Light Detection and Ranging (LiDAR), and cameras, were deliberate towards improving their ability to be more accurate, dependable, and affordable. Such sensors are no longer optional for the much-needed accuracy but a necessity.

VANETs are key software solutions to the IVSCN, which are based on vehicle-to-vehicle (V2V) (et al., 2024) and vehicle-to-infrastructure (V2I) (Xie et al., 2023) communication. These are the onboard units (OBUs),

which are devices used in both transport and road infrastructure and are instrumental in the creation of a complex communication infrastructure (Ganeshkumar & Kumar, 2021). Their contributions to applications such as self-driving cars and real-time traffic control have become critical, as they provide a platform for gathering and sharing information related to vehicles, general traffic situations, and environmental factors. The visions of IVSCNs are multifaceted, with safety being the cornerstone, but they are also about improving traffic efficiency, reducing air pollution, and providing ease and comfort for road users. Through the development of alerts for approaching hazards, clustering of traffic, and restriction of needless fuel expenditure, IVSCNs are important for the progression of environmentally friendly and functioning modern transport networks. Nevertheless, IVSCN design might end up as a complicated and puzzling task that is marred by many challenging factors such as scalability, security, privacy, and interoperability among various coordinated fields of work involving different types of technologies or equipment manufacturers, as well as robustness and reliability under fluctuating conditions.

While the limits of advanced communication systems are still being revealed, the prospects of this technology are thrilling, a future envisioned as one where cars talk to each other and the road network, which ultimately leads to the emergence of cleaner, safer, and more efficient roads.

As technologies continue to improve, edge computing has become a critical element for sensors added to the infrastructure of IVSCNs, which can significantly decrease latency, a major factor in developing highly automated vehicles, and artificial intelligence (AI) can be used for predictive traffic management and autonomous vehicle decision-making. These early achievements, along with world actions by the IEEE and 3GPP(3rd Generation Partnership Project) (Pan & Kao, 2024), and efforts by government agents globally, represent the pooled and focused efforts towards making the Internet of Vehicularly Connected Networks as best as possible. It may be further noted that this particular trend in transportation is not just about adopting such novel technologies but also affects our perception of the road journey's interaction, thus creating fully automated and interconnected vehicular transportation.

The latest research surveys on IVSCN emphasize that the field is of vital importance and is still undergoing rapid advancements, which focus on different areas such as vehicular communications and integration of sensor networks – specifically on the evolution of network architectures, leveraging smart technology solutions such as AI

and Blockchain, and laying emphasis on communication protocols to mobilize vehicle functions.

One remarkable survey (Huang et al., 2024) is concerned with the design features and implications of technology for next-generation video streaming in automotive networks. This study has shown that multimedia from advancements such as 360° and volumetric videos in the coming years, with run-of-the-mill network capabilities, will not meet the requirements of such heavy demand applications in vehicles that are in the process of evolution and development, exerting relentless pressure on next-generation wireless communication technologies.

Another extensive study (Huang et al., 2024) investigated the use of LiDAR-aided channel modelling for vehicular networks in the context of the evolution of protocols and how AI improves communication protocols and network management, thereby reducing costs and increasing the security of the communication process. This study was also intended to determine how AI could help integrate 6G technologies with vehicular networks and improve their performance to a higher level.

Furthermore, the role of intelligent vehicular networks in providing seamless and secure communication among vehicles was discussed in (Muslam, 2024). Such studies have stressed the crucial role of secure and scalable network architectures in providing diverse vehicular communications, from accident prevention to traffic management. Such surveys collectively offer an extensive visualization of the present status and future perspectives of the IVSCN, with a comparable look at technology breakthroughs aimed at reshaping vehicle communication.

Our contribution is a well-organized description of vehicular and sensor communication and an overview of its various technologies and protocols. This research is significant because it provides a structured and comprehensive overview of the current state of IVSCN in modern communications, making it easier for researchers at all levels to plan for the direction of their future studies. We conducted this review by searching the IEEE Xplore database for publications. Research publications that met the criteria were published between 2015 and 2024. The remaining content of this paper is organized as follows. Section 2 describes the background of vehicular communications, and Section 3 provides an overview of the performance models of electric vehicles (EVs). The vehicular network communication protocols are explained in Section 4, and Section 5 discusses routing protocols. The navigation system is described in Section 6. Finally, the conclusion is presented in Section 7.

Technological background

Sensors technologies

Sensor technologies play a central role in the creation of intelligent vehicular networks and ADAS (Advanced Driver Assistance Systems). Radar sensors are popular because they can produce effective results at all times and under all circumstances. These sensors transmit ultrasonic waves and determine the distance and velocities of objects in their surrounding environment by calculating the time required for the echoes to return to the sensors. Thus, the distance, speed, and angle can be measured. This capability is basic for higher-order functions, such as collision avoidance, adaptive cruise control, lane change observation, and parking support systems. Apart from increasing the level of safety during traffic jams, radar sensors also minimize the fatigue resulting from high-speed driving. However, because of certain resolution problems in radar sensors, they are paired with LiDAR and cameras to improve their reliability and functionality (Wang et al., 2020). Other recent works (De Silva et al., 2018; Wang et al., 2023) have discussed the use of radar with other types of sensors, such as LiDAR and cameras, to improve the overall system performance and robustness. For instance, the author of a study (Yeong et al., 2021) described LiDAR and Radar technologies for integration in self-driving cars and the importance of multi-sensor integration to provide an accurate perception of the environment. They focused on various types of sensors, elaborating on their advantages and disadvantages, and characterized methods of data fusion.

LiDAR sensors are vital for the real-time operation of vehicular cyber-physical systems and contribute to the high-definition 3D map creation required for ADS (Autonomous Driving Systems) (Lim & Tuladhar, 2019). LiDAR sensors use lasers to send signals and then receive light waves, created by the change in amplitude, wavelength, or frequency reflected from surfaces, thereby providing accurate information to avoid obstacles, determine a path, and analyze detailed maps. These sensors are useful in pedestrian detection, lane assistance, and adaptive cruise control. Even though LiDAR sensors are greatly influenced by weather conditions, they are very useful in forming a complete sensor setup, thus serving as key inputs for perceiving the surrounding environment and enabling safe and efficient car operation.

Cameras are also essential components that capture the high-resolution scene information needed for safety and navigation systems. They obtain pictures and videos of the surroundings of the vehicle and are used in other functions, such as lane departure and pedestrian detection. The information obtained through these sensors is useful for identifying objects, classifying scenes, and

colour segmentation, which are important factors in the decision-making of complex logic algorithms in automobiles. While radar and LiDAR sensors detect the distance and shapes of objects, high-resolution cameras provide additional layers of depth and detail for the vehicle's interactive environment, thereby enabling the vehicle to think correctly for safe driving. The study (De Silva et al., 2018) aimed to integrate LiDAR and camera inputs to enhance object recognition and categorization. They showed that it is possible to improve the accuracy and stability of autonomous systems by integrating specific sensors, thereby improving the performance of self-driving cars in complicated driving scenarios. However, the combination of LiDAR with radar and camera inputs is crucial for achieving complete environment perception to ensure safe self-driving cars.

Ultrasonic sensors are widely implemented in automated vehicle systems and are accurate and best suited for low-speed and short-range scenarios. These use ultrasonic waves to determine the distance to obstacles by sensing the reflected ultrasonic waves, which can be useful for assisting in parking and detecting blind spots. These are critical for low-speed collision avoidance systems as well as for improving the safety of automobiles.

Vehicles and sensors have come a long way, and the integration strategies that encompass vehicular and sensor networks are quite distinct from the conventional forms of integrated sensing and communication systems (ISAC). IVSCNs must incorporate ultra-low latency, real-time data processing, and data fusion ability, especially for safety-critical operations such as collision avoidance or autonomous vehicles. Owing to the geographical mobility of vehicular environments and the high rates of topology changes, communication protocols must be dynamic and highly adaptive. Similarly, IVSCNs must grow to cover millions of cars, with each being built by a different manufacturer and communicating with a unique infrastructure.

These networks employ efficient protocols along with environmentally-friendly routing strategies to support the aspects of security, traffic control, management, and concern for the environment by enabling the real-time detection of different types of risks and potential threats, dynamic traffic signal control, and nature-efficient energy management. When several sensors are combined with AI analysis, the improvement in accuracy and security is a potential basis for creating ITS. Figure 1 shows the positioning of sensors in automated vehicles and the areas covered by these sensors.

Collecting and processing data from a wide range of sensors, mainly radar, LiDAR cameras, and ultrasonic sensors, are fundamental in smart vehicular networks to ensure a greater understanding of the interaction with

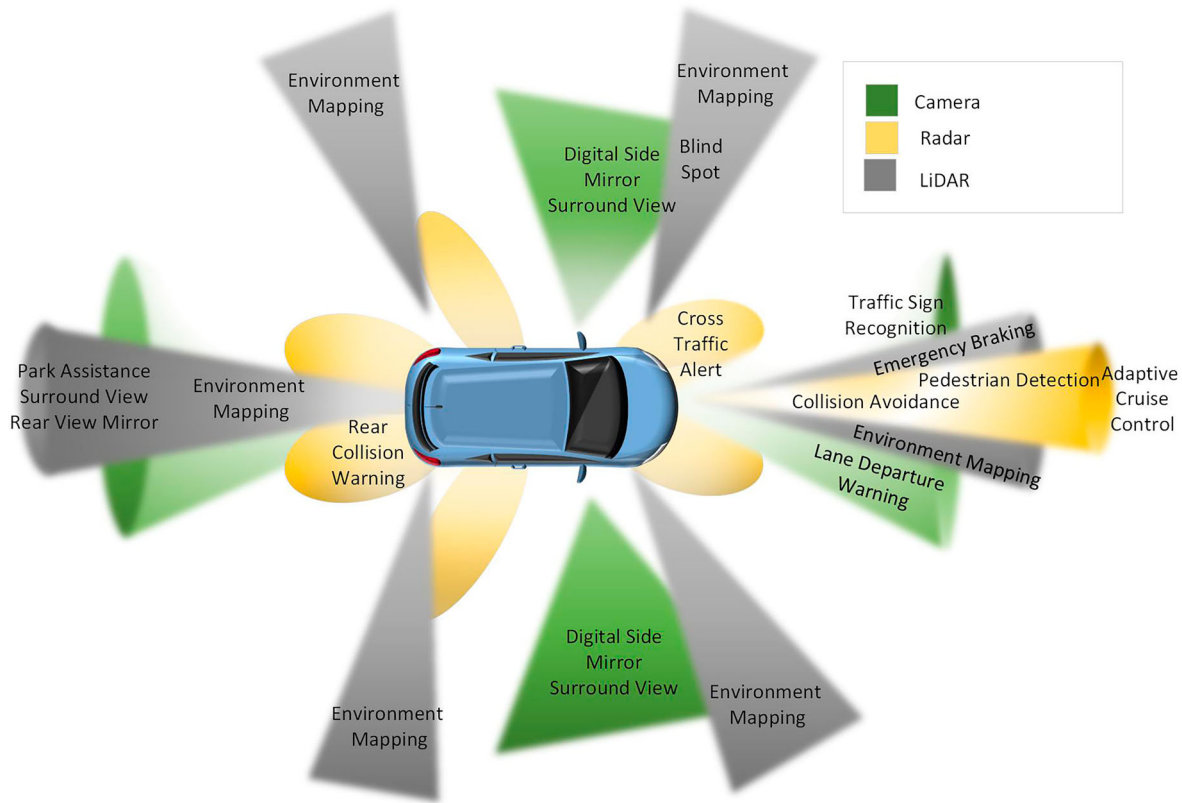


Figure 1. Position of sensors in an automated vehicle.

the environment of the vehicle. Radar sensors obtain information on distance, speed, and angle, which are all related to objects using radio waves, thereby enhancing the coverage of objects for systems such as collision avoidance and speed modulation. LiDAR sensors improve the accuracy of 3-dimensional scans by generating laser beams that create maps of surroundings with the highest resolution needed for a more complex environment understanding and precise driving. Sensors that work like human eyes capture large images of traffic points, which enables advanced image processing approaches such as object detection, sign identification, and lane detection. They are of great importance in the processes of decision-makers and supply the data in the type needed to add to the data from the other sensors. Ultrasonic sensors, which mainly prevail for low-speed sensors, are fundamental to the monitoring of blind spots and alerting and parking aids by determining the echo of the ultrasonic waves reflected from nearby objects.

The sensor data processing involves multi-layer processing; raw data fusion is the first step, which combines the data from multiple sensor sources to create a coherent picture, and only secondary processing is performed (Pati et al., 2024; Yeong et al., 2021). This is accomplished by feature-level fusion, in which features extracted from individual sensor data streams

are integrated (Hasanujjaman et al., 2023). In addition, decision-level fusion combines acquired data and extracted versatile elements to make intelligent decisions for vehicle management and operation (De Silva et al., 2018). The Maximum Heading Similarity (MHS) metrics for sensor fusion are expressed as:

$$MHS_{f1} = \sum_{j=1}^n \max s(\tilde{r}) \quad (1)$$

In Equation (1), $s(\tilde{r})$ is the estimated fusion value of the autonomous vehicle sensors. The Average Detection Precision (ADP), $s(\tilde{r})$, for cars, sensors are expressed as:

$$s(\tilde{r}) = \hat{p}(c_c, i) \cup \hat{p}(l_c, i) \cup \hat{p}(r_c, i) \quad (2)$$

where $\hat{p}(c_c, i)$, $\hat{p}(l_c, i)$, and $\hat{p}(r_c, i)$ are the individual AV detection precisions of the car camera, LiDAR, and RADAR, respectively. Sensor Data Integration and Time synchronization of diverse sensors are accomplished by powerful onboard processing units of computers with high computing capabilities, and edge computing is executed to minimize the time delay. Such a network has powerful information processing capabilities because AI will function on our roads. It is reliable, stable, and independent, regardless of road conditions.

However, it is difficult to compile information from several sensors without any challenges. The integration

and coordination of several data streams lead to technical complexity. Sensor function and integration are also impaired by factors such as fogging, rainfall, and snow. Some of these challenges include cost and scalability; for example, the cost of developing certain sensors, such as LiDAR, cannot support the implementation of such technologies in all classes of vehicles. They must be highly reliable and safe for all possible operations, as any failure in the operation of a sensor can impact the entire system. In addition, larger quantities of data raised data privacy and data security questions related to data management and storage.

Communication technologies

Vehicular communications have the feature of being an evolutionary step in transportation, which works as a link between electronic technologies for automobiles and a digital interface. A dynamic area outstanding in this sector, Vehicle-to-Everything (V2X) communications (Kumar & Mikkili, 2024) is a vehicle-related digital interaction sphere covering the entire spectrum of V2X communications (Khezri et al., 2022). These connections enable the creation of ITS, which in turn helps improve road safety, promote traffic efficiency, and create a better environment by reducing traffic congestion and emissions. The core of vehicular communications involves various types, each serving a unique role: V2V (Muslam, 2024), V2I (Khan et al., 2022), Vehicle-to-Network (V2N) (Ma et al., 2022), and Vehicle-to-Pedestrian (V2P) (Kabil et al., 2022; Malik et al., 2020) communication. This results in the real-time transmittance of critical information such as speed, direction, and location, which allows vehicles to anticipate potential danger, improve traffic management, and minimize pedestrian risks.

Vehicular communications, a cornerstone of intelligent transportation systems, are underpinned by two critical technologies: DSRC and C-V2X (Khalid et al., 2024). Wireless communications between vehicles, such as the DSRC, specially dedicated to automotive functions, allow for immediate and reliable messaging over short distances, and finally dispense with the need for cellular networks. This technique is based on the IEEE 802.11p standard (Nai et al., 2017), which is an adapted form of traditional Wi-Fi protocols to meet the needs of vehicular environments and offers multiplexing of data streams quickly to other neighbours in the area of vehicular coverage (Xue et al., 2024). Studies (Nai et al., 2017; Yin et al., 2004) have provided performance comparisons and laboratory simulations of DSRC that explain the better performance in urban environments. However, owing to the aforementioned drawbacks of DSRC, including its small deployment range and scalability issues, research on

other technologies must commence. The IEEE 1609 Wireless Access in Vehicular Environments (WAVE) suite standard dictates security procedures, network, and transport services alongside multi-channel operation, enabling the whole framework to take the shape of a reliable and secure surveillance infrastructure across different channels (Clancy et al., 2024; IEEE, 2023).

However, it relies on the well-built infrastructure of the present cellular networks comprising 4G LTE and 5G upgrades to facilitate transport communications to a wider extent (Agiwal et al., 2021). C-V2X, which is governed by the 3GPP standards, introduces two different modes. The direct mode of the PC5 interface is essential for establishing direct V2V and V2I communications and laying the foundations for split-second safety applications such as crash avoidance and traffic flow optimization. In addition, the device-supported network-facilitated model still relies on the out-of-date UU interface of the cellular networks to connect to broader online networks for accessing cloud-based services such as navigation, telematics, and entertainment options. C-V2X specifications, such as the Released 14 Project and Release 14B (McCarthy et al., 2021), are created that involve V2V and V2I exchanges to take place without network coverage. This competence may be expected to increase with the succeeding versions alongside the integration of 5G New Radio technologies, which render them promising cost-saving benefits on delay, reliability, support of a denser network of connected devices, and reduced downtime (Kumar et al., 2024). Similarly, in (Khalid et al., 2024) and (Moradi-Pari et al., 2023), the authors compared DSRC and C-V2X, where the latter was found to provide better performance in terms of coverage, data rate, and network integration. These studies mentioned that the integration of C-V2X with a 5G network is possible and may provide advantages, such as support for high traffic density and more intensive V2X communication. It is expected that the new transition from 5G-NR V2X will eliminate the aforementioned challenges and pave the way for another fulfilling future of V2V self-driving.

In short, DSRC and C-V2X constitute a technological base that combines the main forces of modern road V2X communication systems. Each method has its advantages. With its reliable, low-delay transduction for short-range applications, such as parking, emergency services, and traffic signals, DSRC provides communication in the form of a wireless network (Okpok et al., 2024; Qi et al., 2020). Meanwhile, C-V2X technology substitutes the traditional V2X communication system through the utilization of existing and expanding cellular networks, which in turn becomes a gateway for numerous applications that seek to make intelligent transportation systems safer,

more efficient, and more convenience-oriented (Flowers et al., 2024; Mir et al., 2020).

Network design and architecture

Architectural frameworks

The formulation of an intelligent vehicle and sensor communication network architecture acts as a design template for the integration of technologies and the building of a reliable system. The considered structures involve the concept of network detail and component build-ups, such as sensors, data collectors, communicators, and interfaces. The framework normally consists of a sensor perception layer for acquiring data, a communication protocol for a network layer, a security and integrity transport layer to ensure safe and sound data operation, as well as data for application-specific sensors, navigation, safety features, and an automatic operation issues layer. The design should not only be scalable, adaptable, and robust enough to handle the gang of future technological advancements and growing traffic demands, but also purposefully created.

Network topologies in vehicular communication

The network topology determines the layout of the nodes among the network participants, which can be vehicles, RSUs, and central nodes. Typical topologies of in-vehicle networking feature stars, meshes, and hybrid configurations (Gheorghiu et al., 2017). The star topology typically consists of a central node, typically known as an RSU, to which vehicles are connected through direct links. This architecture reduces the complexity of setup and management as it is centralized, making it efficient in environments with static, low-mobility nodes (Hegde et al., 2013). The main drawback though is the lack of redundancy of the whole network that can be affected by a single point of failure. Moreover, scalability becomes a problem as the network continues to increase, which might cause the central node to become a bottleneck.

In addition, the mesh topology enables V2V communication without relying on a central node. This operation makes the network highly reliable and fault-tolerant because the collapse of one node does not affect the others. Scalability is another advantage of mesh networks, which also enables a larger number of vehicles. Although this poses some difficulties in the form of coordination and management of data, it can become increasingly complex as the network grows. Furthermore, the possibility of increasing the number of communication hops can increase the latency in large setups (Basha et al., 2022; Chakraborty & Nandi, 2011). The hybrid

topology is designed with the sole purpose of emulating the strengths of the star and mesh topologies and enhancing the overall network performance. It provides the advantage of flexibility by combining centralized and decentralized communications according to requirements, improving fault tolerance without sacrificing any centralized control at the same time. This pattern seeks to achieve a balance between latency and throughput by selecting the best route and managing the network under real-time conditions and requirements. The hybrid topology is beneficial but complex to implement and might lead to a need for more resources for the system to be effective and obtain proper supervision and management (Zhang et al., 2019).

Edge and cloud computing in vehicular networks

The interoperability between edge and cloud computing plays a crucial role in managing intelligent vehicular networks by facilitating computations, as shown in Figure 2. Edge computing operates near the source of data, for example, in vehicles or roadside units, which leads to a reduction in latency, which is particularly important for real-time automotive applications. Being close to the data minimizes latency for critical pathways such as collision avoidance and additionally carries the benefit of saving bandwidth, as there are no extra loads on the central server to handle, and also resolves network congestion (Verschoor et al., 2023; Wang et al., 2018). However, edge computing technology has its limitations because it is dependent on the processing power and storage capacity of edge devices, and it can be difficult to maintain the widespread edge infrastructure, which can be costly. The paper (Hou et al., 2016) explains in detail the practical implications of vehicular fog computing, also referred to as edge computing, where the vehicles themselves are used as computing infrastructures and share data. The use of this principle shortens the time it takes for data to be processed and for decisions to be made within smart transportation systems, which include emergency response and traffic analytics. Although edge computing offers operational efficiency and low latency, the computing capability of vehicles is limited and poses issues of data security and privacy owing to its distributed architecture.

In addition, cloud computing locates the data processing itself in remote data centres, thereby offering extensive computational power and space for storing enormous amounts of data used for analyzing and managing accumulated data. This encourages cloud computing to be best suited for non-real-time tasks that require extensive data analysis, such as traffic pattern analysis and vehicle diagnostics (Naseer Qureshi et al., 2018). However,

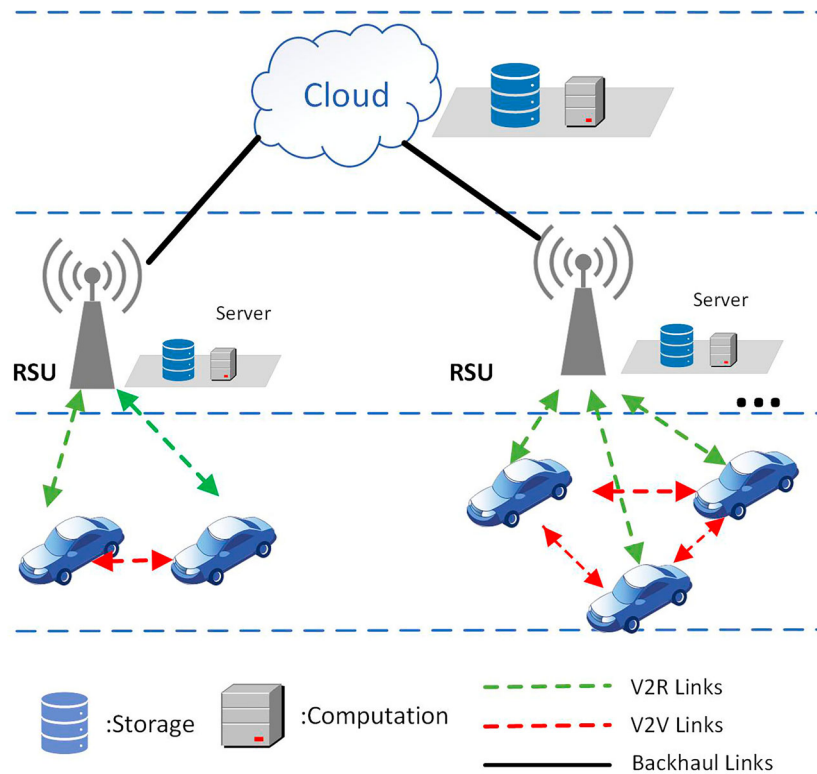


Figure 2. Edge and cloud computing in vehicular networks.

there are also some major disadvantages, including high latency, because data needs to travel a greater distance and dependency upon fast, reliable internet connectivity, which is essential for any time-sensitive application. In (Zhang et al., 2017), the authors intend to examine cloud computing and how it works in vehicular networks with a considerable focus on preemptive off-loading, where vehicular data tasks are migrated from edge devices to a centralized cloud to deal with the load effectively. The case study demonstrated that integration with cloud computing would bring about significant growth in computational capabilities and, as a result, improve network management in large-scale data processing tasks. Even though the system is highly dependent on a stable network connection, higher latency might be a problem in safety applications where timely responses are very important.

Decisions between edge and cloud computing for vehicle networks generally depend on the specific requirements of the application. Edge computing is beneficial for real-time and safety-critical applications because it provides low latency and instant processing. In parallel, the cloud is preferred for massive operations that involve extensive data analysis and is not time-sensitive because of its computational robustness and economies of scale. Over time, hybrid methods that can combine both edge and cloud computing systems have become

increasingly popular, as they effectively use the benefits of each approach. These designs make the edge enable the initial processing of data in real-time, while more exhaustive data analysis and storage are performed in the cloud. This hybrid not only carries the network responsiveness and data rate but also makes the networks more trustworthy and scalable for in-vehicle communication (Jiang et al., 2022).

Introducing fully autonomous vehicles in a vehicular network brings potential issues that must be considered. Salhaoui (2021) and Farooqi et al. (2022) investigated the applicability of edge and cloud computing to real-time data analytics and decision-making for autonomous systems. Farooqi et al. (2022) introduced a vehicular fog computing model based on vehicles as mobile computation nodes to minimize communication delays. This approach extends the possibilities of scaling and fault tolerance of vehicular networks, and therefore, corresponds to the need for autonomous driving.

Furthermore, Biswas and Wang (2023) and Shruti and Rani (2024) highlighted the significance of incorporating features such as AI and machine learning (ML) in vehicular network architectures. Biswas and Wang (2023) specifically note that edge computing helps self-driving cars, in which latency-sensitive tasks, such as ACC or automatic braking, must be performed rapidly. Shruti and Rani (2024) presented a fog computing model for smart cities

that delineated the optimization of traffic systems by handling vehicle sensor-based data collectively with urban structure data in real-time.

Using these references as a single entity provides extensive knowledge of the current status of intelligent vehicular and sensor communication networks, including technologies, methodologies, and applications. Incorporating a detailed analysis and discussion into this new portion, the section provides an enhanced understanding of the issues and prospects in the field, specifically the development of self-driving cars.

Performance models

The purpose of performance models in IVSCNs is not only to include engineering parameters such as latency, throughput, and reliability but also to include operational parameters such as the fuel consumption and emissions of a vehicle, the consumption of energy to maintain the network, and the costs of communication. Therefore, this model can serve as an essential tool in the evaluation of viability, economic efficiency, and the level of environmental influence of IVSCNs.

Performance models are required to be efficient and effective in the sense that they should function properly in the network environment and at the same time cause minimal harm to the environment. Seo et al. (2021) and Jereb et al. (2021) highlighted other models that attempt to predict fuel consumption and emissions as a function of traffic conditions and vehicle movement. These models are essential for the assessment of ITS from an environmental impact perspective. For instance, Jereb et al. (2021) used engine thermal models and vehicle dynamics to estimate CO₂ emissions and compare different types of traffic signal control, which would be useful in understanding the effect on the environment.

Some of the most recent studies (Fan et al., 2024) and (Ahn et al., 2023) analyzed the energy requirements of connected and autonomous vehicles (CAVs). The studies proved that careful incorporation of CAVs into urban traffic schemes can potentially reduce total fuel consumption and emissions when eco-routing and adaptive speed control are involved. In this context, the results presented herein reveal that intelligent vehicular networks can significantly contribute to improving sustainability at the global level.

Several studies have targeted solutions, such as vehicle fuel consumption and emission model creation. Their reliance on macroscopic fuel consumption and emission models has been proposed in (Heni et al., 2023). However, they did not analyze the transient variations in the speed and acceleration of a vehicle. To outperform this constraint, models of microscopic fuel consumption and

emissions have been proposed in (ProQuest, 2024; Sun et al., 2021), where the fuel consumption and emissions of a vehicle can be predicted in approximately a second. Figure 3 shows a summary of the links between transportation, fuel consumption, and emission models.

A new module was proposed in Bento et al. (2019), in which the varying CO₂ emissions of traffic passing through intersections were measured using an engine's thermal model and a vehicle's longitudinal dynamics. To compute the necessary engine torque, the first step is to calculate the engine rotational speed, ω_{eng} , using the wheel radius, r_{wheel} , and the gear ratio, γ :

$$\omega_{eng} = \frac{\bar{v}_{veh}}{r_{wheel}} \cdot \gamma \quad (3)$$

where, the gear ratio γ is a function of $\bar{v}_{veh}$. To achieve the determined vehicle speed, the engine must apply the following torque T_{eng} to the crankshaft:

$$T_{eng} = \frac{F_{trac}}{\omega_{eng} / \bar{v}_{veh}} \quad (4)$$

For the engine's torque and efficiency characteristics, the Willans approximation was used:

$$\rho_{me} = \epsilon \cdot \rho_{mf} - \rho_{me0} \quad (5)$$

where ρ_{me} is the brake mean effective pressure, ρ_{mf} is the fuel mean effective pressure, ϵ is the indicated engine efficiency, and ρ_{me0} represents all mechanical friction and pumping losses. The Willans model is preferred because of its fewer computational requirements and reliable engine modelling for real engine behaviour. This kind of evaluation is affected by the energetic dynamics of propulsion components on the full-speed cycle, which should meet the desired dynamic profile. It has the advantage of low computing requirements and the use of algorithms that generate detailed fuel demands and emissions as much as possible for any speed profile. Further (Li et al., 2023), research was centred on the factual compounds of fuel, speed, and exhaust on CAVs in daily urban traffic to determine the total cost of fuel consumption on the road under different maximum passenger rates. By conducting a simulation of the energy utilization of the vehicles, the technology creates total energy waste, which is followed by an examination to determine the increase that occurs after the implementation of CAVs. By having numerous examples on roads with electric and fuel vehicles, the energy consumption and emissions as a collective are improved by this scale as the number of AVs increases. Considering the heavy reliance of fully EVs on electricity, electric cars have therefore become more of an investment in infrastructure for cities than cars themselves. eq. (6) can be used to calculate the CO₂ emissions

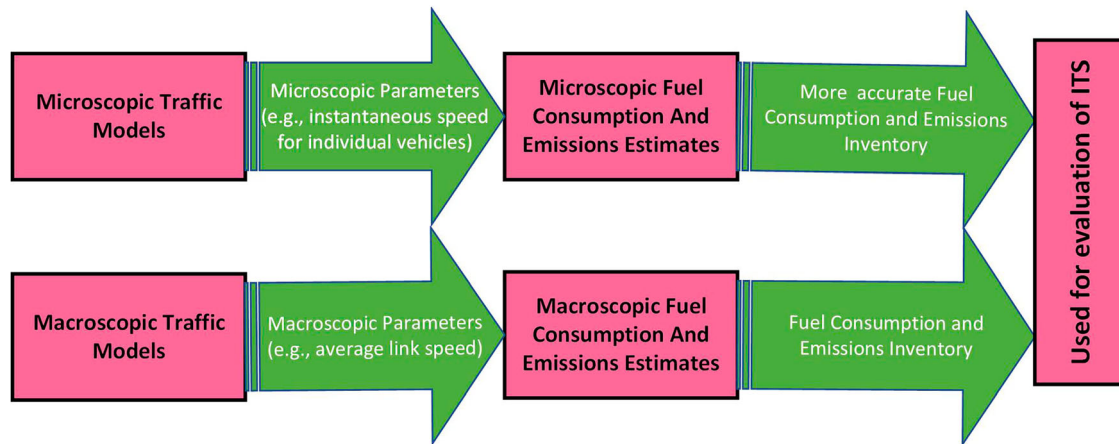


Figure 3. Summary of the link between transportation and fuel consumption and emission models (Alsabaan et al., 2010).

from EVs in terms of uniformly used electricity.

$$E_{(CO_2)} = 1 * 10^3 . E_{elec} . \epsilon \quad (6)$$

where $E_{(CO_2)}$ is the emission of CO_2 (mg/km), E_{elec} is the electricity consumption (Wh/km), and ϵ is the conversion coefficient (0.785).

In wireless communication, the level of power mobilization differs from system to system. For example, although cellular base stations have relatively loose power usage requirements, handsets and other devices must be in full power-saving mode, depending on the circumstances of the user; otherwise, their transmission will interfere. However, there is a kind of bias towards the efficiency of power in mobile ad hoc and wireless sensor networks, which typically enjoy a more sustainable energy supply than vehicle networks for their communications (Devika et al., 2023). The deployment and increasing utilization of EVs and Hybrid Electric vehicles (HEVs) is a spark that drives changes in vehicular communication protocols, basically to allow the optimization of power consumption in EVs, considering that EVs are powered by limited resources. Even though current vehicles have a massive amount of research on fuel efficiency, power usage improvements in EVs via VANETs have rarely been considered. In addition, RSUs are given a strategic location that offers an opportunity to improve network communication efficiency and power usage. Using the network strategies employed by cellular networks for the optimal placement of RSUs (Vieira et al., 2023) can also result in increased efficiency of message delivery, reduction in latency, and extended coverage range, all of which contribute to energy saving, thus citing this area as a dynamic avenue for research and development to reinforce EV communication systems (Feng et al., 2023; Sagaama et al., 2023).

The communication costs were also included in the performance models. Energy saving is a significant concern in wireless communication, and several energy-saving protocols have been proposed. The issue of energy consumption has always been secondary in traditional auto networks and is now a key factor, as green protocols are being implemented to interconnect with other standards, such as IEEE 802. 11p. Research (IEEE, 2024) evaluated the energy profiles of V2R and V2V systems by evaluating their transmission, reception, stand-by, and other energy consumptions. Although the estimation of communication costs is not usually the purpose of these studies, they reveal that energy consumption perspectives play a significant role in connected vehicle networks (Min et al., 2023).

Vehicular network communication protocols

Network architecture

For vehicular networks, the design of the networking structure undoubtedly has a centralized role, specifically when adopting sensing techniques onboard. Decentralized networks can be considered, where all vehicles are connected directly through V2V communication (Ji et al., 2020). This approach is complex because it does not depend on the use of prebuilt infrastructure and learns to manage heterogeneous car movement dynamics by preventing stop-and-go type scenarios and reducing speed and acceleration optimization, thus wasting less fuel and energy while achieving enhanced travel efficiency (Zheng et al., 2015). However, the introduction of permanent infrastructure into the network may increase communication reliability but incur a higher cost. To maintain equilibrium, a hybrid product is frequently proposed, in which the infrastructure is rolled out in a targeted manner. Such an architecture provides cars with the option to

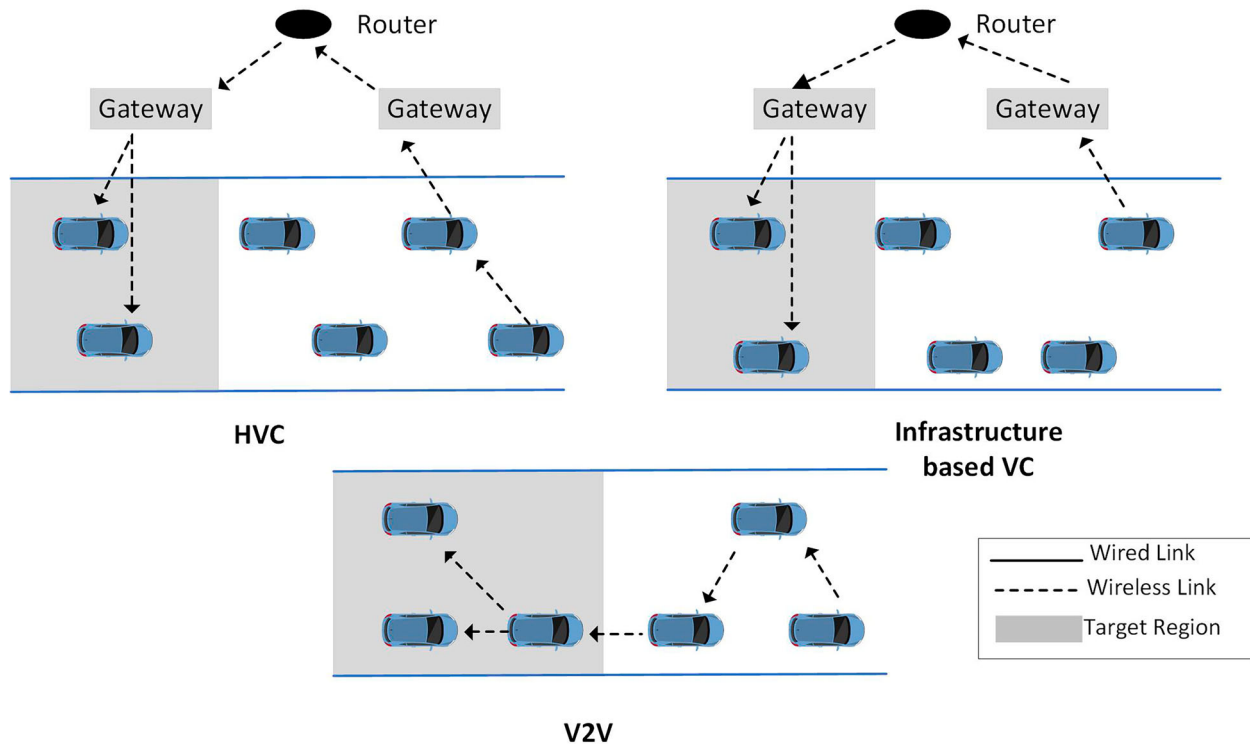


Figure 4. Network architectures in vehicular networks.

either directly contact the infrastructure or utilize other cars for transmitting information, and assures the utility of this special model as it adapts to different requirements and circumstances. Figure 4 shows the different network architectures of vehicular networks.

Traffic scenario

We encountered three types of traffic scenarios: Traffic Light signal (TLS) scenarios, accident scenarios, and congestion scenarios. For instance, vehicles travelling along road sections controlled by TLS are a result of behaviours that increase unnecessary fuel consumption, thereby impacting both fuel efficiency and environmental sustainability. Such factors include the stop-and-go scenario, where vehicles are stopped at a red light and only succeed in advancing as the light turns green, which leads to an increase in fuel consumption and emissions due to idling and the anticipation of the next acceleration (Ngo et al., 2019). The other energy-demanding action is when the vehicles increase their speed unnecessarily, even before an approaching TLS, instead of lowering it to a moderate speed. Moreover, it may be that cars using TLS without stopping, such as those approaching the speed, require aggressive acceleration after the intersection. They prefer to reach maximum speed faster to avoid traffic instead of losing time accelerating at an appropriate speed that minimizes unnecessary acceleration

post-intersection. Such non-standard behaviours imply an area of investigation where adaptive control systems or intelligent strategies are available that ensure the performance of vehicles in approaching TLSs efficiently, conserving energy, minimizing emissions, and optimizing traffic (Olaverri-Monreal et al., 2018).

Highway accidents may occur for different reasons and have various consequences, so it is crucial to establish appropriate safety measures and crisis management measures (Biswas et al., 2006). Rear-end collisions or hitting the vehicle in front due to distracted driving, often caused by slow-moving traffic, standing traffic, or rush-hour traffic, stopping suddenly, or driving too close to the car in front of them (Ye et al., 2008). Red light running or non-compliance with stop signs leading to intersection accidents influences side impacts, head-on collisions, and encounters with pedestrians and bicyclists (Azimi et al., 2013). Lane departure crashes, associated with inattention, fatigue, or poor road conditions, occur when vehicles run along the edge of a lane with other cars or non-road objects (Motro et al., 2016). Thus, head-on collisions, where one vehicle is travelling in the opposite direction, which is the most lethal type of crash, often occur when a driver enters an oncoming lane or makes a wrong-way entry on the highway (Chang et al., 2019). Collisions between cars and non-motorists are common during turns, at high speeds, and when a car driver is distracted; they result in some of the most severe or fatal incidents

(Jing et al., 2017). Usually, they occur as single-vehicle accidents due to distracted driving, poor road conditions, or to avoid an animal, which results in contact with barriers or roll-overs (Jeong et al., 2021). In conditions such as poor visibility, multi-vehicle pile-ups are not unusual and present unique difficulties because of the large number of vehicles and people affected (Mohammed et al., 2019). These include Poor Road conditions, such as fog and snowy or rainy grounds; weak signs also play a role in causing accidents because they minimize the visibility and holding of cars (Vahdat-Nejad et al., 2016). Features such as advanced braking systems, lane assist features, and adaptive cruise control, followed by vehicular communication technologies (V2X), help prevent these accidents as they involve direct communication between the vehicles and the infrastructure to ensure that drivers are aware of the risks.

Congestion on roads, one of the most common challenges in urban areas, is when the demand for using the roads is higher than the road capacity, causing traffic to slow or stop. A resounding illustration of this problem is that travel during rush hour plus car accidents, road construction, or adverse effects of weather, becomes very frequent (Paranjothi et al., 2020). Gridlock not only creates more headaches for commuters and extends travel times for them, but also increases fuel consumption, elevates emissions of greenhouse gases and pollutants, and reduces air quality. In addition, it adds to economic losses because prolonged transport of goods ensues, and the punctuality of the public transportation system decreases. As a result, aftermarket makers have started to join forces and develop various ITS that use Vehicular Network Communication technologies, such as V2V or V2I. These systems aim to improve the transport schedule in two ways: they provide real-time traffic information to the passengers and authorities, which will enable them to control traffic signals adaptively, and they introduce alternative routes and modes of transportation to bypass the congested parts of the roads and improve general road network efficiency (de Sousa et al., 2020).

Generalized system model

According to the literature, vehicular communication systems communicate with each other using many models and architectural structures, as shown in Figure 5. This points to the development of a single model that captures all elements needed while addressing internal communication dynamics. Ordinarily, road networks consist of both urban locations and suburban areas, including TLSs and RSUs, at strategic intersections and intervals to establish connections. RSUs enhance the network by

operating with short-range wireless connectivity that allows them, as well as additional equipment, to transmit data and information around the system (Benzagouta et al., 2023). By placing sensors along the road length, the roadway network captures valuable traffic stats like vehicle speed, which include information, such as fuel consumption, and pollution level (Mao et al., 2023). The network architecture categorizes nodes as mobile (vehicles) and stationary (TLSs and RSUs), supporting diverse communication pathways, the most important of which are V2V, V2I, and I2I (infrastructure-to-infrastructure).

A central component of the vehicular communication network is the OBUs and TLSs, which are directly responsible for maintaining connectivity between vehicles, antenna access, ad hoc routing, and traffic control, including congestion control. Every OBU and TLS is equipped with a wireless communication module for short-range communication. Along with Vehicles and TLSs, Application Units (AUs) embrace the functionality of various applications, including GPS and electronic maps, with full vehicle navigation. The essential components of OBUs and AUs include the following:

- Short-range wireless devices: Radio communication, such as IEEE 802.11 and IEEE 802.11p for point-to-point channel access, is involved.
- Driver Interface: Provides crucial feedback to the driver; the information is in the form of audio or visual cues.
- Navigation System: Composed of a GPS receiver, digital maps, and route selection software to help find a better route to move one's current location to the chosen destination.
- TLS Controller: The function is not only term control, but also the operation of TLS, which differs from static to dynamic with the adjustment of real-time traffic control.

According to the Car-to-Car Communication Consortium (C2C-CC), a layered structure with details of application and transport layers will be the basis of the proposed protocol stack, and there are discussions on how to specify these protocols (Agrawal et al., 2023). This incorporates ad hoc routing, which facilitates efficient communication between vehicles, as well as between vehicles and RSU or TLS, with the IEEE 802.11p standard at the MAC and PHY layers, implying that the architecture is designed in a complex but structured manner to avoid system failure.

Data dissemination

Data dissemination protocols in Vehicle Network Communication based on the speed, mobility, and variable

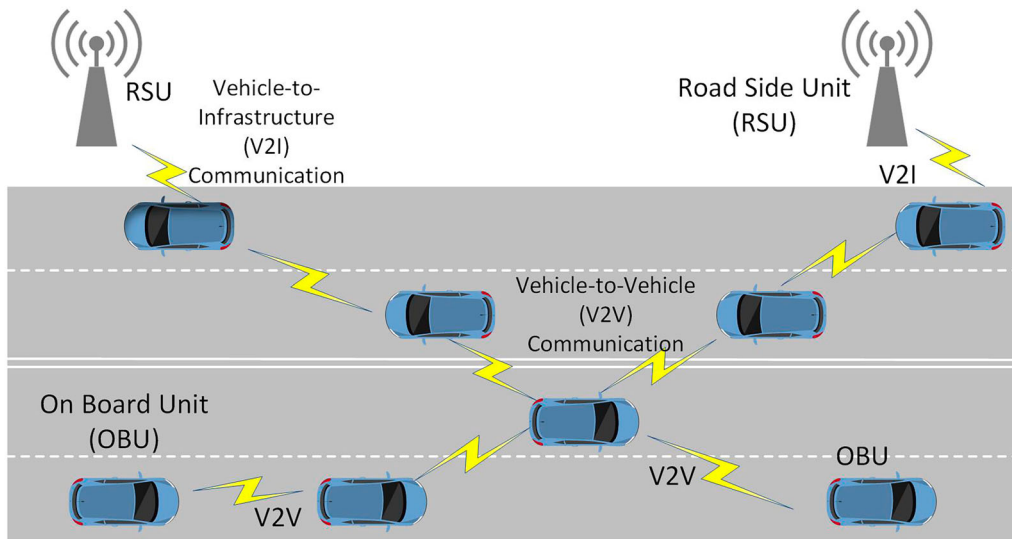


Figure 5. General system model of vehicular networks.

density that the vehicles encounter must be implemented to enable the efficient and reliable transmission of information between vehicles and road infrastructure, which is typical of a vehicular environment (Gupta et al., 2020). The procedures include broadcast-based means for the release of emergency alerts covering the entire population, multicast-based means for information sharing targeted to specific groups of interest, geo-cast-based for location-specific messages, and cluster-based means to administer the vehicular networks that are of the highest density, and by splitting them into these clusters (Gopalan et al., 2023). To handle issues such as extreme topology changes, network partitioning, and the necessity of secure communication, adaptive techniques, storage, and carry forward mechanisms, and encryption as well as authentication, are very serious measures. Developed with processing power, reliability, and efficiency capabilities in mind, these protocols are involved in a wide range of services, from emergency notifications to controlling traffic and entertainment services, thereby becoming an integral part of an intelligent transport system, which is characterized by safer roads and better traffic flow.

Routing protocols

The routing protocols for vehicular networks and sensor communication allow data packets to travel through networks, from one vehicle to another, or from vehicles to infrastructure, taking into account high vehicle mobility, rapidly changing network topology, and variable node density (Yogarayan et al., 2020). In a heterogeneous group of protocols, geographic routing protocols concentrate on incorporating geographical information

into the decision-making process to make smart routing decisions (Din et al., 2020). Every node determines its place, neighbours' positions, and the goal location to find the best path to send the data packets further (Qureshi et al., 2020; Tambe et al., 2024). The distance metric was calculated using Equation (7).

$$Distance = \max \left[\log \left(\frac{SD}{SN} \right) \right] \quad (7)$$

where SD is the Source to a Destination node, and SN is the Source to Neighbour Node (SN). A specific example worth mentioning is the Greedy Perimeter Stateless Routing (GPSR) protocol that provides the closest neighbour to the destination with the shortest path; this route is efficient (Albattah et al., 2022).

Moreover, Topology-based Routing Protocols seek to ride on existing connections to draw the most feasible routes through which the data may travel (Sehrawat & Chawla, 2023). These categories of protocols are subdivided into two categories: proactive protocols that carry out constant updates on driving routes throughout the network and reactive protocols such as Ad-hoc On-demand Distance Vector (AODV) (Taterh et al., 2020), which dynamically finds routes based on the state of the network – particularly useful in vehicular context because of its high mobility (Shrivastava & Vishwamitra, 2021).

Broadcast Routing Protocols are crucial in situations where information needs to be transmitted widely, including the dissemination of emergency alerts, implementation of strategies to deal with broadcast storms, and data duplication, mostly through probabilistic and node selection methods (Shah & Kasbe, 2021). In the case of complex and dense vehicular networks, Cluster-Based

Table 1. Routing protocols for reducing fuel consumption.

Type of routing protocol	Main focus	Typical applications
Eco-routing protocols (Priya et al., 2022)	Lessens fuel consumption as well as carbon emissions.	Ideal for everyday driving scenarios where having a minimal negative effect on the environment is critical, such as in urban and suburban traffic.
Traffic-aware routing protocols (Gillani et al., 2022)	Adjusts routes based on traffic patterns encountered while driving.	Use these in complex intersections that are getting congested, to improve the waiting time and ultimately cut off fuel usage.
Predictive routing protocols (Khalid Diaa et al., 2023)	Employs historical and real-time data.	This system could be fit for scheduling tasks in traffic-congested areas by forecasting congestion and recommending alternative routes.
Speed optimization protocols (Valayapalayam Kittusamy et al., 2022)	Recommends optimal travel speeds	Interestingly, on the highways and long trips, keeping a fast speed is the right choice that can decrease fuel consumption a lot.
Cooperative adaptive cruise control (CACC) (Liu & Wang, 2021)	Optimizes the speed of individual vehicles.	Primarily used on the highways to keep the required following distance and speed, thus reducing the chances of unnecessary stops and starts.

Routing Protocols improve node management and scalability by grouping vehicles into clusters and appointing a leader for each cluster ([Divya et al., 2022](#); [Naeem et al., 2023](#)). They play a part in coordinating within and between clusters. Simultaneously, they handle communication tasks locally and globally. Delay-Tolerant Network (DTN) protocols are significant for gray areas in which connectivity is not completely reliable ([Singh & Pamula, 2021](#)). This paradigm enables data caching within vehicles so that they can store data until a network connection is available, which is essential, especially in remote parts of vehicular networks ([Wu et al., 2020](#)).

Similarly, Multicast Routing Protocols are appropriate for applications that deliver messages to multiple specific destinations simultaneously ([Babu & Parthiban, 2022](#)). However, the dynamic nature of the vehicular network generates unique new problems; however, these protocols are fundamental for negotiations on the synchrony of vehicle maneuvers, turning all vehicular communications into a highly complex process. Multi-protocol routing protocols for both vehicular and sensor networks exist, and they were developed to meet the unique requirements of a dynamic vehicular environment. Table 1 presents precisely designed protocols in vehicular communication networks, providing an overview of different types of routing protocols for reducing fuel consumption, with an overview of their main focus and typical applications.

Data management and analytics

Data acquisition and fusion techniques

The process of collecting data in an intelligent vehicular environment induces the capture of data from several sensors, including radar, LiDAR, cameras, and ultrasonic devices. Sensors of each type contribute to a comprehensive understanding of the car's environment as well as its current functional state. Fusion methods unify this diversity of data from different sensors into a composite model

of the environment. This integration occurs at various levels, starting with the raw data fusing technique of combining the same content types of data and proceeding to the feature level fusion techniques of combining feature levels extracted for analysis from different sensors, so we reach a final level of the decision-level fusion technique that aggregates outputs from independent data processing systems to achieve a comprehensive decision ([Butt et al., 2022](#)). Effective data fusion not only improves the accuracy of modelling the lathering water environment but also helps this system be more reliable by the partial of individual transducer sensors so that the single sensors are not left alone.

In this section, we discuss several sensor data fusion methods that are generally applied to vehicular networks. The Kalman filter, typically employed in statistical algorithms, performs the task of uniting data from various sensors while actively minimizing the mean squared error (used extensively in the presence of Gaussian noise). This strategy is normally used in vehicle navigation and tracking systems ([Farak, 2021](#)), where it fuses data from the GPS, accelerometers, and gyroscopes into the linear dynamics of the system to provide the best possible estimation results. Although it works well enough in linear systems, it fails to perform well in non-linear systems unless the Kalman filters are revised, such as the Extended Kalman Filter or the Unscented Kalman Filter.

Alternatively, Particle Filters (or Sequential Monte Carlo methods) provide a robust solution to non-linear and non-Gaussian environments, making them suitable for adverse driving conditions typical in adaptive cruise control systems. Although it is more adaptable and sturdier than Kalman Filters, Particle Filters demand high computational capacity, which may limit their implementation in real-time ([Ying et al., 2021](#)).

The third method is fuzzy logic fusion, which applies fuzzy sets and logical rules to the data and successfully handles the uncertainties and imprecision in sensor

readings. This technique is widely used in driver support systems that include fusing data from visual, radar, and ultrasonic sensors for obstacle detection and classification (Abbasi et al., 2022). Although fuzzy Logic is effective in the uncertainty of fusion, its performance based on the design of fuzzy rules and membership functions depends on it.

Numerous IEEE studies have focused on this area, predominantly highlighting the progress and hurdles in sensor data fusion for automotive networks. As an example, the approach suggested in Shahian Jahromi et al. (2019) looks at how it is possible to balance the computational cost and reliability of data explored in real-time sensor data fusion. The goal is to achieve sufficient performance for autonomous vehicles by providing a system that is capable of interpreting complex sensory inputs and that is effective in terms of performance. Another contribution is research (Wang et al., 2023) in 2022, which focuses on the creation of a fusion approach to join radar and LiDAR data to increase object detection accuracy under different driving conditions. This study concentrates on the issue of overcoming the flaws of each sensor type using a contrived fusion algorithm that provides greater accuracy of detection and system reliability.

Real-time data processing

Real-time data processing is a necessary function in vehicular networks that enables timely decisions that affect safety and efficiency. It involves making control functions based on sensor information within milliseconds to achieve the best functioning of the vehicle. Real-time processing leads to functionalities such as adaptive cruise control, automatic braking, and dynamic route guidance, which are based on current traffic conditions. The implementation of these systems requires a high amount of computational resources, which are usually imposed by both onboard processors and edge devices to minimize the delay (Liu et al., 2019). An architecture that can handle big data quickly should be put in place as well as a fault-tolerant mechanism that is meant to avoid cases where the failure could lead to safety risks.

As mentioned in (Lu et al., 2022), this study proposes the construction of a scalable and highly efficient architecture for data processing with features such as massiveness and suitability for autonomous cars. The main goal of the current study is to improve the accuracy of real-time decision-making in automated driving by reducing latency and improving the efficiency of data processing systems. In another study (Ning et al., 2019), it was demonstrated how vehicle data used in conjunction with urban infrastructure enhances traffic flow and public safety. This paper discusses a new method to organize

large volumes of sensor data in real-time to create an efficient system that can reduce response times and improve data accuracy by utilizing fog computing technologies.

Data analytics in vehicular networks

Data analytics, serving as the core of vehicular networks, provides the ability to convert the information collected by vehicle sensors into operational goods, thus enabling better traffic management, vehicle efficiency, and safety. Information gathering and data processing of a more complex nature, where knowledge is drawn from various sources, such as Radar, LiDAR, cameras, and ultrasonic sensors, which may be installed either on vehicles or in the traffic infrastructure. Utilizing intelligent algorithms ML (Zidi et al., 2024), these data are a primary source for the prediction of traffic patterns, optimization of routes, advancement in vehicle autonomy, and increased safety measures, including predictive maintenance and real-time detection of hazards.

A visible influence can be expressed in several IEEE publications, which provide examples of the progress and implementation of data analytics engineering in traffic networks. For instance, (Zhang et al., 2011) is dedicated to researching the application of data analytics in traffic system optimization and traffic congestion reduction. The author of this article explains real-time data processing and ML technology for gaining better traffic forecasts and management strategies. Another research article (Pandharipande et al., 2023) addressed the fact that ML algorithms are integral to processing the sensor data needed for the implementation of automation features, such as smart adaptive cruise control and collision avoidance systems.

Case studies and practical implementations

In recent years, the application of communication systems in intelligent vehicles has led to the realization of many benefits and challenges arising from these novel technologies. An interesting case is presented in one of the papers on the IEEE Network (Ehlers et al., 2017), which is related to the use of Cooperative Intelligent Transportation Systems (C-ITS) in Europe. It is mentioned that C-ITS relies on V2X communication to enable vehicles as well as infrastructure to share data; therefore, traffic becomes safer and more efficient. To illustrate, the aforementioned research demonstrates that the adoption of C-ITS results in fewer accidents and traffic jams as indirect outcomes. Nevertheless, the research study only targets European regions, meaning it could not provide an accurate representation of the traffic conditions in global markets. As such, a 2018 study (Giust et al., 2018) indicated

that networks operating across metropolitan areas using edge computing for processing data from vehicle sensors were being performed in real-time to lower latency and optimize traffic management, thus allowing for faster response to changing traffic conditions. Some of the main strengths are that it addresses the issue of processing data in real-time, as well as the issue of interconnectivity. However, problems arise in the actual implementation of a network structure when applied to a larger number of connected automobiles.

In Hilmani et al. (2018), the article explains the development of an intelligent parking system that aims to solve the parking problem in cities with high population density, which has been worsened by rural-urban migration. The system uses wireless sensor networks (WSN) and Internet of Things (IoT) technologies and employs a self-organization algorithm that is built to work decently on every kind of car park. It improves energy management, increases the sensor's lifetime, and simplifies the allocation and exploration of free parking lots. This mobile app provides affordable, fast, and sufficiently good services. Moreover, the system utilizes wireless sensor networks and IoT technologies, improves energy management, lengthens sensor node life, and facilitates the parking process by providing speedy and effective searches for open parking lots. The main contribution is how effective it works in urban areas; however, its high implementation costs could make it impractical in some areas.

Furthermore, vehicles installed as infrastructure blocks for fog computing were introduced in Hou et al. (2016). Edge computing utilization in smart transit employs widely used vehicles to develop processing capabilities at the network edge to quickly solve the issue of time-sensitive applications, such as emergency vehicle routing. The adoption of vehicles as nodes of a fog computing system architecture ensures not only a delay reduction in urban settings but also an increase in the scalability and resilience of vehicular networks. In contrast, the network has to confront technicalities, such as the uneven distribution of vehicles and variations within the computing capabilities of vehicles, which could influence the conformability and uniformity of the network's performance. Implementation of this novel approach demonstrates a departure from conventional network structures through the process of data processing decentralization, which implies many possible increases in the efficiency of intelligent transportation systems.

In 2022, research (Zhang et al., 2023) highlighted the implementation of an integrated edge-cloud architecture with a high emphasis on ITS. This complex architecture fully combines end, edge, and cloud computing

features to guide adaptive data traversal and processing, thus overcoming the shortcomings of bandwidth and latency. The main functionality of this architecture is ML and forecasting along the vehicle trajectory to disperse the safety notifications that minimize network load and data transmission. The application of this strategy has proven to be effective in dense motor vehicle systems, ensuring minimal interference and improving the effectiveness of the network. However, the architecture leads to a lot of extra complexity, particularly the need for highly precise models. This eventually becomes a problem for implementation and future operations. In this study, authors provide a good example for future development in the field of network design, promoting the virtues of layered computing architecture in vehicular networks.

The survey (Torres-Alvarado et al., 2022) on the subject of hardware architectures for quality detection in software-defined vehicular networks (SDVN) was carried out using ML algorithms such as decision trees and Multi-Layer perceptrons. This study aimed to identify the correct channels that would improve communication efficiency and reliability within vehicular networks. Such resulting algorithms, coupled with the facility of dynamic adaptation to the ever-changing environment, allowed the study to make clear communication channel choices depending on their quality. Nonetheless, the requirement of specific equipment and the application of sophisticated ML algorithms increase the expense of the system and system complexity. This thorough investigation significantly contributes to the question of using ML for immediate communication in a constrained environment, thus providing exceptionally useful inputs for designing communication channels in dynamic networks.

In addition to pinpointing the real use cases, these case studies are more than just a confirmation of the practical implementations of intelligent vehicular networks. They also shed light on their operational performance, providing a reference that orients future deployments in other regions and specific contexts. Table 2 compares this study with previous studies, emphasizing their contributions, methodologies, key findings, strengths, and weaknesses.

Altogether, these case studies contribute to the development of intelligent vehicular networks by showcasing real-world use cases and the value of integrating AI, including intelligent traffic control and management, safety and security solutions, parking space availability, and real-time analytics. These aspects range from V2X communication and edge computing to IoT and machine learning, where, despite providing strong outcomes, they have challenges.

Table 2. Comparison of existing papers.

Study	Contributions	Methodology	Key findings	Strengths	Weaknesses
Ehlers et al. (2017)	Implementation of C-ITS	V2X communication and its practical applications	With C-ITS, there will be fewer accidents and faster commuting time.	Offers practical benefits of C-ITS throughout Europe	Restricted to situations in Europe
Giust et al. (2018)	Optimizing traffic flow in real-time with edge computing	Combining edge computing with real-time data processing	Optimizes traffic flow and reduces delays	Offers better traffic management and response	Interoperability and scalability issues
Hilmani et al. (2018)	Developed a smart parking system using WSN and the Internet of Things	Using WSN, the Internet of Things, and algorithms for self-organization	Increases the lifespan of sensors, energy management, and parking lot efficiency	Effectively addresses parking issues in cities	Extremely high implementation costs in highly populated regions
Hou et al. (2016)	For fog-computing systems used vehicles as infrastructure block	Computational edge technologies for smart transportation	Benefits urban areas by increasing scalability and decreasing latency	Using vehicles as edge nodes in a novel way	Performance is impacted by an uneven vehicle distribution.
Zhang et al. (2023)	Edge-to-edge cloud architecture for ITS integration	Combining machine learning with cloud, edge, and end computing	Provides advantages to dense networks via enhanced data traversal, decreased network load, and enhanced efficiency.	Effectively manage congested transport networks	Increasing intricacy and need for accurate modelling
Torres-Alvarado et al. (2022)	Configurations for assessing the quality of SDVNs	Use of MLP and Decision Trees, two methods for ML	Increases the communication's reliability and effectiveness	Can be easily adjusted to suit different network conditions	Additional complexity and cost to the system
This paper	Gives a comprehensive analysis of IVSCN	Structure review of recent Advancements and practical implementations of IVSCN	Advancement in communication protocols	Technical assessment of sensor technologies provides ideas for further investigation.	May lack precise implementation details

For example, studies of both C-ITS and edge computing are concerned with real-time data processing and network capacity. On the other hand, the intelligent parking system and fog computing research topics are slightly closer to the given problem, as they are more related to energy consumption and resource usage. The integrated edge-cloud structure and hardware structure investigation focuses on the challenges of managing large-scale data and maintaining communication.

Collectively, the outcomes of these studies can help address critical industry issues, including the coordination of traffic lights, the routing of emergency vehicles, and parking management in urban areas. Highway safety analysis can provide support for policy-making to reduce traffic congestion and encourage safer roadways. Areas for future research include the improvement of utilizing AI for real-time operation, improvements in sensors, and better protocols.

However, there are some limitations to modern approaches, such as integration and compatibility issues between technologies. Future work will explore improved data fusion algorithms, more flexible learning models, and extensible network architectures. Further research on low-cost architectures for mass deployment and solutions to data privacy and security challenges will be important for the development of intelligent vehicular networks.

Applications of intelligent vehicular networks

Intelligent vehicular networks have been set to bring about significant changes in traditional transportation systems through their use of smart sensors, robust communication mechanisms, and efficient data analytics. These networks cover different aspects of the road and can achieve an array of applications that improve vehicle functionality, increase road safety, and decrease traffic congestion.

Enhanced road safety and collision avoidance

One of the main uses of intelligent vehicular networks is to improve road safety by decreasing the occurrence and severity of accidents. With the use of sensors such as radar, LiDAR, and cameras, which can identify risks immediately, cars can detect obstacles and reckless behaviours of pedestrians, cyclists, and other vehicles. The V2V and V2I communication networks can immediately transmit this information to vehicles, instructing them with automatic braking and evasive action mechanisms, helping to prevent collisions on time by improving decision-making.

Efficient traffic management

Intelligent transportation systems that rely on interconnected autonomous vehicle networks to manage and optimize traffic flow in urban and suburban areas are also crucial. Through the use of commercial vehicles and roadside sensor data, traffic management systems are supposed to monitor and analyze traffic continuously to adjust signal phases in real-time to reduce congestion and optimize traffic patterns. Therefore, not only does this commuting help in reducing vehicle emissions, but also in minimizing idling, stop-and-go driving conditions stop the emission levels of vehicles.

Support for autonomous vehicles

An intelligent vehicular network is the backbone of autonomous vehicle operations, so that these automated, high-skilled systems can perform securely and properly. Autonomous vehicles obtain updated data from the sensor array and external environment systems to proceed without direct human control. Finally, these vehicles have the capacity for significant savings in gasoline consumption, oil prices, and income for the citizens. Such networks provide venues for vehicles with the capability of automation to have an ongoing conversation among themselves, which keeps them updated on road conditions and traffic changes, among other relevant information that enables intelligent decisions.

Real-time environmental monitoring

However, intelligent vehicular networks play an indirect role in transport service provisioning. In addition to this rather obvious application, these networks find another use in environmental monitoring. The purpose of a vehicle can be altered from its conventional one to collect data as a mobile sensing station; variables such as air quality, temperature, and other environmental parameters are some factors that contribute to data collection. These data could be helpful in the follow-up of any environmental trend, for public health measures, and can be used by other intelligent responses to climate change, for example, adjusting traffic flow in the event of low air quality.

Emergency response enhancement

Intelligent vehicular networks improve the efficiency of emergency response strategies by ensuring that emergency vehicles are properly routed immediately. A V2I communication channel enables road authorities to assign emergency services a priority rank that ensures

that traffic signals are synchronized to let them go through, and other vehicles on the road are notified to make way. This will help reduce the number of responses and increase the safety of emergency responders and the general public.

Smart parking solutions

The search for parking spots is an overwhelming task in densely packed cities. Solutions to this problem are based on intelligent vehicular networks, in which drivers are redirected to places with available, convenient parking spots based on real data. This application is designed to reduce the time spent by drivers searching for parking, with the result in easing traffic congestion as well as emissions from cars that are circling.

Infotainment and connectivity

Lastly, these networks not only offer infotainment and connectivity features that make the ride even more delightful but also bring out the kind of convenience that increases the driver's comfort levels. Starting from streaming media and real-time traffic updates to vehicle diagnostics and communication, the driver and passengers can, now on board, a journey that combines and works together with telematics, entirely enjoy a connected ride.

Future research directions

The field of IVSCNs is rapidly developing, providing many possibilities for future research. These guidelines not only aim to solve today's problems but also include searching possibilities, which could change the vision of these infrastructures, especially for metropolitan and marginal infrastructures. Some key research directions are as follows:

Enhanced sensor technologies

The next step in future research could be an attempt to improve the quality of sensors by either sensing them more precisely or developing them for wider-ranging and affordable uses. Modifying the sensors such that they can provide more precise and meaningful data under all situations, even in a dynamic environment of intelligent vehicular networks, will significantly improve their network capabilities. In addition, the use of multimodal sensor techniques will be beneficial for the accurate and complete perception of the environment, which is essential for the safe operation of autopilot units.

Early research was based on the importance of sensor integration, especially radar, LiDAR, and camera types, to enhance vehicle perception systems. In 2023 (Wang et al., 2024), multisensory fusion techniques were utilized to improve the accuracy of object detection in complex environments. Nevertheless, these systems encounter a high number of processing loads and conflicting data from sensors, particularly under difficult weather conditions, as well as loss of sight from the front. Future research needs to be more oriented towards lighter but effective data fusion algorithms, such as Kalman filtering, particle filtering, and Bayesian networks, for real-time data fusion and the buildup of conflict resolution mechanisms that improve the quality of environmental sensing under diverse physical conditions.

Machine learning and artificial intelligence

The use of ML and AI in vehicular networks tends to increase the research on algorithms that could advance without the intervention of humans, which learn, adapt, and induce on their own to constantly changing conditions and new data. This is demonstrated by employing predictive traffic management models, real-time decision-making systems for driverless transport, and anomaly detection systems for repair and safety.

Previous research (Haddad et al., 2022) used deep reinforcement learning to optimize traffic signal operations in real-time. In this case, the systems still require a large amount of data for training and computational resources, which can be impractical in dynamic environments. This challenge can be solved by developing a lightweight ML model that requires fewer computational resources and can adapt to changes in both the environment and the object. Techniques such as model pruning, knowledge distillation, and transfer learning can be exploited to reduce the size of models without any deterioration or even improvement in their performance. This can be beneficial in reducing the need for large amounts of training data and extensive computational facilities.

By emphasizing the use of low-cost sensor technologies and enhancing ML techniques, IVSCNs can be made simpler and more affordable. Reduced computing needs can be achieved through sensor fusion, which will improve the accuracy of affordable microelectromechanical sensors. TinyML and feature selection approaches are lightweight models that can contribute to this improvement. Edge computing also lessens the need for expensive cloud infrastructure, which in turn lowers operational expenses and delays. Scalability can be further enhanced by efficient communication methods like Narrowband-LoT and Long Range Wide Area Network (LoRaWAN), which reduce power consumption and running costs.

These strategies ensure a very efficient and cost-effective approach.

Another study (Zhang et al., 2018) employed convolutional neural networks (CNNs) to classify network traffic. A CNN can be effective for static scenes, but its level of performance can be low in non-stationary network conditions, such as those that appear on roads, because they contain scenes unseen by the learning algorithm. When considering channel degradation in non-stationary network situations, the creation of adaptive ML models with self-learning capabilities is very important. The models can learn and adapt to the changing environment by themselves without requalification to address the changing environment. Technologies such as continuous digital learning or reinforcement learning that can improve the knowledge base in real-time are very useful in such situations.

In (Ramamoorthy & Velu, 2024), the researchers discussed clustering algorithms for VANETS. The algorithms can optimize the management of the overall communication process and simultaneously direct communication. Nevertheless, they face problems with high mobility; this phenomenon results in dynamic topology changes that often lead to re-clustering and high overhead. One of the most critical research directions is the development of clustering algorithms that can address these changes more effectively. Provided by the complex algorithms that can respond to changes in the network topology quickly, probably using predictive analytics to foresee the changes and reroute ahead of time, the necessity to frequently re-cluster and communication overhead might be minimized.

Communication technologies and network protocols

In the past, vehicular communication systems mostly used several key technologies and protocols to ensure that data transfer was carried out effectively and securely between vehicles and infrastructure. The area of vehicular communication technology has been profoundly transformed by recent innovations, encompassing Cellular-V2X (C-V2X) and 5G New Radio-V2X (NR-V2X). C-V2X can operate in modes with self-organizing properties so that vehicles can autonomously handle transmission resources and management; however, the technology is not very reliable in dense traffic and high speed. NR-V2X is a novel 5G concept that provides higher data rates and ensures direct communication between vehicles. Meanwhile, both wireless technologies struggle with high-density conditions and require upgrades to link capacity and resource management to deal with the optimal mix of vehicular connectivity and a variety of communication needs (ElGhanam et al.,

2021). To eliminate these drawbacks, future research should focus on several aspects, such as the exploration of algorithms that can efficiently high-density vehicular environments to prevent overload and optimize the use of available resources. Studies on technologies that provide higher link capacity and a greater number of addresses without deteriorating the level of service. To investigate communication techniques such as network slicing, dynamic spectrum allocation, and multi-access edge computing (MEC) to enhance reliability, performance, and diversity in different contexts and circumstances.

Cybersecurity measures

Recent enhancements in cybersecurity include the incorporation of federated learning models that facilitate cyberattack detection across vehicular networks. This technique provides a module for sharing the analytics of data collected from different vehicles on one platform, while its privacy remains secure. Blockchain technology provides for the integrity and non-repudiation of transactions and communication within a vehicular network. This helps eliminate the need for a centralized communication system in V2X communication services and provides a decentralized approach. It is necessary to consider the use of more advanced encryption methods between vehicles and infrastructure to make data security more efficient. This implies the use of quantum-resistant algorithms to stay ahead of time for all possible technological changes. Hence, current models that use diverse cybersecurity solutions are inherently restricted to deeply congested vehicular areas where communication between multiple nodes needs to be quick and frequent. Integrating high-level encryption procedures and blockchain tools into vehicular communication systems makes them more complex and expensive. Incorporating additional security processes, such as those based on cryptography or more than one testing method, may lead to delays. Such delays are unfavourable in systems demanding data interchange in real-time mode.

Overcoming such challenges for future research will involve identifying lightweight and scalable protocols that can simultaneously handle the high rate and low delay of vehicular networks. Building cost-effective cybersecurity solutions that can be easily adopted by all businesses, with minimal financial expenses. In addition, examining techniques such as homomorphic encryption, secure multi-party computation, and zero-knowledge proofs to reduce the latency introduced by security protocols is helpful, as security measures should not affect how the real-time vehicular communication system functions.

Uneven distribution of vehicles

Congestion and inefficient resource utilization can result from an uneven distribution of vehicles in vehicular networks. Optimal network performance can be achieved in real-time with the help of dynamic resource allocation systems, such as adaptive scheduling and load-balancing algorithms. Using predictive methods based on ML, it is feasible to foresee traffic patterns and make proactive adjustments to communication resources. Coordination between vehicles and infrastructure has the potential to increase network efficiency, boost service quality, and ensure continuous connectivity.

Hybrid communication architectures incorporating 5G, edge computing, and satellite networks can be studied in the future to enhance this strategy and ensure continuous service in both urban and rural areas. Decentralized and blockchain-based solutions can improve the reliability and security of sharing vehicle resources. Dynamic route adjustments can be made by AI-powered congestion-aware routing systems to distribute traffic loads more evenly. By taking into account the incentives of various network players, game-theoretic models can be built to optimize decisions regarding the distribution of resources. Digital twin technology improves real-time optimization by simulating and assessing network performance under various vehicle distribution scenarios.

Environmental impact and sustainability

In terms of real-world environmental sustainability, previous research has mainly focused on environment-friendly technologies, such as eco-routing and vehicle electrification. For example, a study from the IEEE Vehicular Networking Conference (VNC) in 2021 (CFP, 2024) showed how vehicle networks and IoT-associated technologies can improve environmental sustainability. In another article (Guo et al., 2015), EcoSky outlined how eco-friendly routing could be utilized for pollution reduction by enriching road networks with environmental data.

Future research may focus on developing smarter and more precise algorithms that are capable of making corrections in routes depending on the environmental factors and reality of the transport situation, to further reduce emissions. Vehicular networks must be interwoven with renewable energy sources, particularly solar and wind, to fuel communication systems and vehicles. Therefore, the entire system is sustainable. Performing LCA (Life Cycle Assessment) during the design and operation of vehicles and sensors helps ensure that the entire cycle, from production to waste, is considered. With further research, it is possible to consider new routing algorithms

that integrate parameters, such as real-time environmental conditions, anticipating disruptions, and the usage of AI.

Conclusions

In conclusion, this comprehensive survey considers the main aspects essential for the development of contemporary vehicular technologies, which is the design of IVSCN. The discussion highlighted various sensors that provide smooth and safe operation of vehicles, in addition to advanced network systems that establish reliable vehicle communication. Performance models were used to analyze fuel and electricity consumption, as well as communication costs. This study delves into the critical analysis of vehicle network communication protocols and navigation systems, with an emphasis on the speed at which data processing is performed and the various scenarios that include traffic light signalling, accidents, and congestion management.

In addition, the paper discusses these technologies' practical implementations and real-world application illustrations of how they were used to improve traffic management, safety, and reduce adverse environmental impacts on the environment. Several types of protocols were given, highlighting their functions in reducing vehicle gas consumption and emissions, with protocols prescribed to be followed based on the scenario and model provided. This study highlighted that more creativity is required for the improvement of routing protocols, navigation systems, and data analytics. This purpose was to supply the fundamental aspects and a crystal-clear vision of the upcoming development of intelligent vehicular networks; therefore, this study will be a point of inspiration for future research and development in this dynamic field. The major contribution of this study is that it provides a valuable database for researchers. It provided new and existing researchers with in-depth knowledge of the things that have been achieved, applications, ongoing research, and what has been done in the area of intelligent vehicular communication.

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Data availability statement

No data were generated in this work.

Authors' contributions

Conceptualization, A.A.A. and H.U.M; Formal analysis, A.M.; Investigation A.M.; Methodology, A.A.A. and H.U.M; Project administration, A.A.A.; Resources, A.A.A. and A.M.; Software, A.A.A. and A.M.; Supervision, A.A.A. and H.U.M; Validation, A.M. and H.U.M; Visualization, A.A.A.; Writing – original draft, A.M. and H.U.M; Writing – review & editing, A.A.A. and A.M. All authors have read and agreed to the published version of the manuscript.

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